Beam Structured Turbo Receiver for HF Skywave Massive MIMO Communications

Linfeng Song*†, Ding Shi†, Xiqi Gao*†, Geoffrey Ye Li‡, and Xiang-Gen Xia§
*National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China
†Purple Mountain Laboratories, Nanjing 211111, China
‡Department of Electrical and Electronic Engineering, Imperial College London, London SW7 2AZ, U.K.
§Departement of Electrical and Computer Engineering, University of Delaware, Newark, DE 19716 USA
Email: songlf@seu.edu.cn, shiding@pmlabs.com.cn, xqgao@seu.edu.cn, geoffrey.li@imperial.ac.uk, xxia@ee.udel.edu

***Abstract*—In this paper, we investigate receiver design for high frequency (HF) skywave massive multiple-input multiple-output (MIMO) communications. We first establish a modified beam based channel model by performing uniform sampling for directional cosine with deterministic sampling interval, where the beam matrix is constructed as discrete Fourier transform (DFT)-based structure. Rooted in the modified beam based channel model, we propose a beam structured turbo receiver (BSTR) involving low-dimensional beam structured signal detection for grouped user terminals (UTs). Then we present efficient implementation of the BSTR by exploiting the structure properties of the beam matrix and the beam domain channel sparsity. Simulation results validate the superior performance of the proposed BSTR with low complexity.**

***Index Terms*—Massive MIMO, HF skywave communications, beam structured turbo receiver.**

I. Introduction

As the demand for global coverage continues to rise [1], high frequency (HF) communications, whose frequency range is 3 to 30 MHz, have emerged as a highly promising technology, allowing for long-range over-the-horizon communication through ionosphere refraction [2]. However, due to the limited spectrum resource and complex ionosphere conditions, the data rates are relatively low in conventional point-to-point HF communications. To significantly enhance the spectrum and energy efficiencies, massive MIMO, a key technology for 5G system and its evolution [3], has been recently introduced in HF skywave communications [4]–[6].

In order to fully leverage the benefits of the HF skywave massive MIMO, there is a considerable demand for receiver design with prominent performance and relatively low computational cost. Linear receivers are widely utilized, where zero-forcing (ZF) receiver and minimum mean-squared error (MMSE) receiver are particularly popular [7], [8], and some approximate methods are proposed to avoid high-dimensional matrix operations of linear receivers [9], [10]. For spatially correlated massive MIMO channels, beam domain sparsity has been employed to design low-complexity receivers [11]–[14]. On account of the small angle spread in HF communications, by exploiting the beam domain sparsity, there are several related works about HF skywave massive MIMO receiver design. In [15], Slepian transform based detectors are derived by utilizing the index set for limited non-zero beams of each user terminal (UT). Space domain and space-frequency domain beam structured detectors are proposed in [16], involving low-dimensional beam domain detector designs. However, the performance of the mentioned receivers for HF skywave massive MIMO is unsatisfactory when the inter-UT interference becomes severe. Originated from the principle of turbo codes [17], turbo receiver has been widely adopted to improve the system performance by iteratively exchanging soft information about the coded bits between a soft-input soft-output (SISO) detector and a SISO decoder [18], [19]. For massive MIMO transmission, it is promising to utilize the beam domain sparsity of massive MIMO channels in turbo receiver design.

In this paper, we investigate the receiver design for HF skywave massive MIMO communications. We propose a modified beam based channel model for HF skywave massive MIMO using sampled steering vectors with deterministic sampling interval for directional cosine. The beam matrix composed by sampled steering vectors exhibits phase-shifted partial discrete Fourier transform (DFT) structure, which is beneficial for low-complexity transmitter and receiver design. Building on the modified beam based channel model, we derive a beam structured turbo receiver (BSTR) by taking advantage of the beam domain channel sparsity. Specifically, we categorize all UTs into finite groups and low-dimensional beam structured signal detection for each group is designed with the MMSE criterion using the a priori information, where the used beams are the union of the non-zero beam sets of the UTs within each group.

II. System Model

A. System Configuration

Consider a long-range HF skywave massive MIMO system with orthogonal frequency division multiplexing (OFDM) modulation. The BS is equipped with M-antenna uniform linear array (ULA), serving U single-antenna UTs. Due to

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the decametric wavelength of the HF waveband, the antenna array aperture of the BS is typically large, resulting in the impracticable of multiple antennas in the elevation direction, different from terrestrial cellular systems.

In contrast to traditional terrestrial cellular communications, HF skywave communication systems require the carrier frequency f_{c} to be adjusted accordingly with the complicated ionospheric environment variations [4]. To address this issue, we define the highest system operating frequency as f_{o} and the wavelength $\lambda_{\mathrm{o}} = c/f_{\mathrm{o}}$, where c is the speed of light. The inter-antenna spacing is set as $d = \lambda_{\mathrm{o}}/2$, which is shorter than half wavelength of carrier different from that widely used in traditional massive MIMO communications.

B. Modified Beam Based Channel Model

For uplink transmission, U complex symbols from different UTs are transmitted over the channel and received at the BS with M-dimensional signal, for each subcarrier of OFDM symbols, which can be expressed by

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{z}, \tag{1}$$

where indexes of OFDM symbol and subcarrier are omitted for brevity, $\mathbf{y} \in \mathbb{C}^M$ is the received signal vector at the BS, $\mathbf{H} = [\mathbf{h}_1, \cdots, \mathbf{h}_U] \in \mathbb{C}^{M\times U}$ is the channel matrix with each column representing the channel response vector for each UT, $\mathbf{x} = [x_1, \cdots, x_U] \in \mathbb{S}^U$ is the transmitted signal vector from U UTs and $\mathbb{S}$ denotes the signal constellation, and $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}_M, \sigma_{\mathrm{z}}\mathbf{I}_M)$ is the additive white Gaussian noise, in which $\mathbf{0}_M$ and $\mathbf{I}_M$ represent M-dimensional all-zero vector and identity matrix, respectively.

To derive the receiver, we now focus on the modelling of the channel. The channel vector $\mathbf{h}_u$ for UT u can be expressed as [15]

$$\mathbf{h}_u = \sum_{p=1}^{P_u} \alpha_{u,p}\mathbf{v}(\Omega_{u,p}), \tag{2}$$

where P_u is the path number from UT u to the BS, $\alpha_{u,p}$ and $\Omega_{u,p}$ are the complex gain and the directional cosine of the p-th path from UT u, respectively. The steering vector $\mathbf{v}(\Omega) \in \mathbb{C}^M$ pointing towards Ω is defined as conjugate centrosymmetric structure with the m-th element being [20]

$$[\mathbf{v}(\Omega)]_m = \frac{1}{\sqrt{M}} \exp\{-\bar{\jmath}\pi f_{\mathrm{c}}\Delta_\tau(2m - M - 1)\Omega\}, \tag{3}$$

where $\Delta_\tau = d/c$.

Owning to the small angle spread of HF skywave communications, the channel can be represented in the beam domain with remarkable sparsity by performing uniform sampling for directional cosine. Specifically, we perform uniform sampling for directional cosine with deterministic interval length

$$\Delta_{\mathrm{an}} = \frac{1}{FMf_{\mathrm{c}}\Delta_\tau} = \frac{2}{FM_{\mathrm{eq}}}, \tag{4}$$

where F is the fine factor, and $M_{\mathrm{eq}} \triangleq Mf_{\mathrm{c}}/f_{\mathrm{o}}$ is the equivalent number of BS antennas. For fixed antenna array aperture $d = \lambda_{\mathrm{o}}/2$ and antenna number M, the angular resolution of the array is proportional to the carrier frequency f_{c}, thus the term M_{eq} signifies the equivalent number of antennas required to achieve the angular resolution corresponding to the half-wavelength array aperture at any given carrier frequency, and FM_{eq} reflects the sampling density. Consider a symmetric partition of the range of directional cosine $[-1, 1)$, we can obtain $A = 2\left\lfloor \Delta_{\mathrm{an}}^{-1} \right\rfloor + 1$ mutually exclusive subsets, and the a-th subset is defined as follows:

$$\mathcal{S}_a \triangleq \begin{cases} [-1, \Omega_a + \Delta_{\mathrm{an}}/2), & a = 1 \\ [\Omega_a - \Delta_{\mathrm{an}}/2, \Omega_a + \Delta_{\mathrm{an}}/2), & 1 < a < A \\ [\Omega_a - \Delta_{\mathrm{an}}/2, 1), & a = A \end{cases}, \tag{5}$$

where $\Omega_a \triangleq \left(a - 1 - \left\lfloor \Delta_{\mathrm{an}}^{-1} \right\rfloor\right)\Delta_{\mathrm{an}}$. Define set $\mathcal{P}_u \triangleq \{\Omega_{u,1}, \cdots, \Omega_{u,P_u}\}$ containing directional cosines of all the paths for UT u, and map the directional cosine in $\mathcal{S}_a$ to the sampled grid point Ω_a. It can be checked that $\Omega_{u,p}$ is mapped to the $a(\Omega_{u,p})$-th sampled directional cosine, where $a(\Omega) \triangleq \lceil \Delta_{\mathrm{an}}^{-1}\Omega \rfloor + \left\lfloor \Delta_{\mathrm{an}}^{-1} \right\rfloor + 1$ is the mapping function.

With the uniform sampling for directional cosine, the space domain channel $\mathbf{h}_u,\ u \in \mathcal{Z}_U^+$ can be approximated as [4]

$$\mathbf{h}_u \approx \sum_{a=1}^{A} \sum_{\Omega_{u,p}\in\mathcal{P}_u\cap\mathcal{S}_a} \alpha_{u,p}\mathbf{v}_a = \mathbf{V}\mathbf{g}_u, \tag{6}$$

where $\mathcal{Z}_N^+ \triangleq \{1, \cdots, N\}$, $\mathbf{v}_a \triangleq \mathbf{v}(\Omega_a) \in \mathbb{C}^M$ is the sampled steering vector and $\mathbf{V} \triangleq [\mathbf{v}_1, \cdots, \mathbf{v}_A] \in \mathbb{C}^{M\times A}$ is termed as the beam matrix, $\mathbf{g}_u \in \mathbb{C}^A$ is the beam domain channel vector with $[\mathbf{g}_u]_a = \sum_{\Omega_{u,p}\in\mathcal{P}_u\cap\mathcal{S}_a} \alpha_{u,p}$. According to the definition, the beam matrix can be expressed as

$$\mathbf{V} = \frac{1}{\sqrt{M}}\mathbf{I}_{M,S}\mathbf{F}_S\mathbf{\Omega}\overline{\mathbf{\Pi}}_S^{S-(A-1)/2}\mathbf{I}_{S,A}, \tag{7}$$

where $S = FM$, $\mathbf{F}_S = [\mathbf{f}_{S,1}, \cdots, \mathbf{f}_{S,S}] \in \mathbb{C}^{S\times S}$ represents S-dimensional DFT matrix, $\mathbf{I}_{N_1,N_2}$ is composed by the first N_1 columns ($N_1 < N_2$) of $\mathbf{I}_{N_2}$ or the first N_2 rows ($N_1 > N_2$) of $\mathbf{I}_{N_1}$. The phase shift matrix is defined as $\mathbf{\Omega} \triangleq \mathrm{diag}\left\{-\mathbf{I}_{S,2S}\mathbf{f}_{2S,M}^*\right\} \in \mathbb{C}^{S\times S}$, in which $\mathrm{diag}\{\mathbf{a}\}$ is the diagonal matrix with $\mathbf{a}$ along its main diagonal. The forward cyclic shift matrix and its variant are respectively denoted by $\mathbf{\Pi}_S \triangleq [\mathbf{e}_2, \mathbf{e}_3, \cdots, \mathbf{e}_S, \mathbf{e}_1] \in \mathbb{R}^{S\times S}$ and $\overline{\mathbf{\Pi}}_S \triangleq [\mathbf{e}_2, \mathbf{e}_3, \cdots, \mathbf{e}_S, -\mathbf{e}_1] \in \mathbb{R}^{S\times S}$, where $\mathbf{e}_i$ denotes the i-th column of identity matrix.

Each sampled steering vector in the channel representation (6) is associated with a physical beam in the space domain, and all UTs utilize an identical set of sampled steering vectors. Moreover, according to (7), the beam matrix can be viewed as partial of the permutation of a phase-shifted DFT matrix, which is superior to the CZT-based beam matrix since the proposed beam matrix has straightforward structure related to the DFT. Therefore, (6) is termed as the modified beam based channel model, which can be written in a compact form

$$\mathbf{H} = \mathbf{V}\mathbf{G}, \tag{8}$$

where $\mathbf{G} \triangleq [\mathbf{g}_1, \cdots, \mathbf{g}_U] \in \mathbb{C}^{A\times U}$ is the the beam domain channel matrix.

On account of the small angle spread and the limited propagation path number of HF skywave channels, the beam domain channel vector shows significant sparsity, i.e., the number of non-zero elements of $\mathbf{g}_u$ is relatively small [4]. The non-zero beam set for UT u is given as $\mathcal{A}_u \triangleq \left\{a \in \mathcal{Z}_A^+ \middle| [\mathbf{g}_u]_a \neq 0\right\}$ with $A_u = |\mathcal{A}_u|$, which can also be obtained by $\mathcal{A}_u = \left\{a \in \mathcal{Z}_A^+ \middle| [\boldsymbol{\omega}_u]_a \neq 0\right\}$, where the beam domain channel coupling vector is defined as

$$\boldsymbol{\omega}_u \triangleq \mathbb{E}\left\{\mathbf{g}_u \odot \mathbf{g}_u^*\right\} \in \mathbb{R}^A, \tag{9}$$

in which $\odot$ is the Hadamard product operator. The beam domain channel coupling vector $\boldsymbol{\omega}_u$ is typically referred to as the beam domain statistical channel state information (CSI) and possesses slow-varying characteristics [4], thus it is efficient to leverage $\boldsymbol{\omega}_u$ to obtain $\mathcal{A}_u$. Moreover, in this paper, we assume that both the beam domain instantaneous CSI $\mathbf{G}$ and the beam domain statistical CSI $\boldsymbol{\omega}_u,\ u \in \mathcal{Z}_U^+$ are available at the BS.

III. Beam Structured Turbo Receiver

A. An Overview of the Turbo Receiver

Typically, turbo receivers are implemented within the bit-interleaved coded modulation (BICM) systems. The original information bitstream undergoes sequential processing through an encoder and an interleaver at each UT. Subsequently, the interleaved bitstream is partitioned into vectors of length N and mapped to constellation points in $\mathbb{S} = \{s_1, \cdots, s_K\}$ using Gray mapping, where $s_k \in \mathbb{C},\ k \in \mathcal{Z}_K^+$, is the constellation point corresponding to the bit sequence $\mathbf{b}_k = [b_{k,1}, \cdots, b_{k,N}]^{\mathrm{T}} \in \mathbb{B}^N$, and $K = 2^N$ is the modulation order, thereby generating the transmitted symbols. The bit sequence mapped to the transmitted symbol $x_u,\ u \in \mathcal{Z}_U^+$, is employed as $\mathbf{q}_u = [q_{u,1}, \cdots, q_{u,N}]^{\mathrm{T}} \in \mathbb{B}^N$, where $x_u \in \mathbb{S}$ and $\mathbf{q}_u \in \{\mathbf{b}_1, \cdots, \mathbf{b}_K\}$. It is natural to assume that the transmitted symbols of all UTs are uniformly and independently drawn from the signal constellation $\mathbb{S}$ at random, and the transmitted power of U symbols are normalized to unit, i.e., $\mathbb{E}\left\{\mathbf{x}\mathbf{x}^{\mathrm{H}}\right\} = \mathbf{I}_U$. At the BS, the received signal is expressed as (1), and a turbo receiver is adopted, enabling the iterative exchange of soft information between the SISO detector and the SISO decoder during the turbo iteration, while the decoded information bits are obtained at the final iteration. Specifically, the iteration process of the SISO detector consists of the following steps:

1) Signal Detection Using the a Priori Information: In the turbo iteration process, the a priori mean and variance of the transmitted signal $\mathbf{x}$ are respectively denoted as

$$\boldsymbol{\mu} \triangleq \mathbb{E}\{\mathbf{x}\} = [\mu_1, \cdots, \mu_U]^{\mathrm{T}} \in \mathbb{C}^U, \tag{10a}$$

$$\boldsymbol{\Sigma} \triangleq \mathrm{Cov}\{\mathbf{x}, \mathbf{x}\} = \mathrm{diag}\{\sigma_1, \cdots, \sigma_U\} \in \mathbb{R}^{U \times U}. \tag{10b}$$

The values of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ are respectively initialized to $\mathbf{0}_U$ and $\mathbf{I}_U$, and are updated through the iteration process. With the a priori information, the signal detection for UT u aims to make an approximation of $p(\mathbf{y}|x_u = s_k),\ k \in \mathcal{Z}_K^+$, for the extrinsic information calculation, where $p(\cdot)$ denotes the probability density function of continuous random variables.

2) The Extrinsic Information Calculation: Based on the received signal $\mathbf{y}$, the log-likelihood ratio (LLR) of the bit $q_{u,i},\ u \in \mathcal{Z}_U^+, i \in \mathcal{Z}_N^+$ is given as

$$\begin{aligned} L(q_{u,i}) &= \ln \frac{P(q_{u,i}=1|\mathbf{y})}{P(q_{u,i}=0|\mathbf{y})} = \ln \frac{\sum_{s_k \in \mathbb{S}_i^1} P(x_u = s_k|\mathbf{y})}{\sum_{s_k \in \mathbb{S}_i^0} P(x_u = s_k|\mathbf{y})} \\ &= \underbrace{\ln \frac{\sum_{s_k \in \mathbb{S}_i^1} p(\mathbf{y}|x_u = s_k) \prod_{j \neq i}^N P(q_{u,j} = b_{k,j})}{\sum_{s_k \in \mathbb{S}_i^0} p(\mathbf{y}|x_u = s_k) \prod_{j \neq i}^N P(q_{u,j} = b_{k,j})}}_{\triangleq L_{\mathrm{e}}(q_{u,i})} \\ &\quad + \ln \frac{P(q_{u,i}=1)}{P(q_{u,i}=0)}, \end{aligned} \tag{11}$$

where $P(\cdot)$ represents the probability distribution of discrete random variables, $\mathbb{S}_i^b \triangleq \{s_k \in \mathbb{S} | b_{k,i} = b\},\ b \in \mathbb{B}, i \in \mathcal{Z}_N^+$, is the symbol set with the i-th mapping bit $b_{k,i}$ of s_k being b. Furthermore, the probability $P(q_{u,i} = b),\ b \in \mathbb{B}$, is calculated by the a priori LLR $L'_{\mathrm{a}}(q_{u,i})$, and is expressed as $P(q_{u,i} = b) = \frac{1}{2}\left(1 + (2b-1)\tanh\left(L'_{\mathrm{a}}(q_{u,i})/2\right)\right)$. The extrinsic LLR $L_{\mathrm{e}}(q_{u,i})$ is utilized for SISO decoding.

3) The a Priori Information Update: After deinterleaving, $L_{\mathrm{e}}(q_{u,i})$ is mapped to $L_{\mathrm{a}}(c_{u,i})$, where $c_{u,i} = \Pi^{-1}(q_{u,i})$ and $\Pi(\cdot)$ is the interleaver. Let $L_{\mathrm{a}}(c_{u,i})$ denote the a priori soft information for SISO decoder's input, which, after the decoding process, yields $L'_{\mathrm{e}}(c_{u,i})$. Subsequently, $L'_{\mathrm{a}}(q_{u,i})$ is obtained after interleaving, which is then employed as the a priori soft information for the detection input. Based on $L'_{\mathrm{a}}(q_{u,i})$, the updated symbol probability is $P(x_u = s_k) = \prod_{i=1}^N \frac{1}{2}\left(1 + (2b_{k,i}-1)\tanh\left(L'_{\mathrm{a}}(q_{u,i})/2\right)\right)$. The mean and variance are respectively updated as $\mu_u = \sum_{k=1}^K s_k P(x_u = s_k)$, $\sigma_u = \sum_{k=1}^K |s_k|^2 P(x_u = s_k) - |\mu_u|^2$. The updated mean and variance of the transmitted symbol is used as the input (10) for the next iteration of detection. Additionally, $L'_{\mathrm{a}}(q_{u,i})$ is adopted for the calculation of $L_{\mathrm{e}}(q_{u,i})$ at the next turbo iteration.

B. Beam Structured Turbo Receiver

We first focus on the signal detection using the a priori information. To develop a low-complexity signal detection approach, we consider linear detection combined with the modified beam based channel model. Before proceeding, we categorize U UTs into L disjoint groups $\mathcal{N}_1, \cdots, \mathcal{N}_L$ with $N_l = |\mathcal{N}_l|,\ l \in \mathcal{Z}_L^+$, and define the UT selection matrix $\mathbf{N}_l \triangleq \mathbf{S}(U, \mathcal{N}_l) \in \mathbb{R}^{U \times N_l},\ l \in \mathcal{Z}_L^+$, where $\mathbf{S}(N, \mathcal{T}) \triangleq [\mathbf{e}_{n_1}, \mathbf{e}_{n_2}, \cdots, \mathbf{e}_{n_T}] \in \mathbb{R}^{N \times T}$ is the selection matrix with $\mathcal{T} = \{n_1, \cdots, n_T\} \subseteq \mathcal{Z}_N^+$, $n_1 < \cdots < n_T$, and $T = |\mathcal{T}|$. Then the detection of $\mathbf{x}$ can be decoupled into L groups, where the detection for the l-th group's transmitted signal $\mathbf{x}_l \triangleq \mathbf{N}_l^{\mathrm{T}}\mathbf{x} \in \mathbb{C}^{N_l}$ can be given as

$$\widehat{\mathbf{x}}_l = \mathbf{R}_l^{\mathrm{H}}\mathbf{y} + \mathbf{d}_l. \tag{12}$$

The MSE for the detection of $\mathbf{x}_l$ is given by $\epsilon_l(\mathbf{R}_l, \mathbf{d}_l) \triangleq \mathbb{E}\left\{\left\|\mathbf{x}_l - \left(\mathbf{R}_l^{\mathrm{H}}\mathbf{y} + \mathbf{d}_l\right)\right\|_2^2\right\}$. In terms of minimizing MSE, the optimal $\mathbf{R}_l$ and $\mathbf{d}_l$ can be obtained by

$$\mathbf{R}_l^{\mathrm{op}}, \mathbf{n}_l^{\mathrm{op}} = \underset{\mathbf{R}_l \in \mathbb{C}^{M \times N_l}, \mathbf{n}_l \in \mathbb{C}^{N_l}}{\mathrm{argmin}} \epsilon_l(\mathbf{R}_l, \mathbf{n}_l), \tag{13}$$

Solving (13) yields the MMSE detector [21]

$$\mathbf{R}_l^{\mathrm{op}} = \mathbf{H}\left(\mathbf{\Sigma}\mathbf{H}^{\mathrm{H}}\mathbf{H} + \sigma_z\mathbf{I}\right)^{-1}\mathbf{\Sigma}\mathbf{N}_l, \tag{14a}$$

$$\mathbf{d}_l^{\mathrm{op}} = \mathbf{N}_l^{\mathrm{T}}\boldsymbol{\mu} - \left(\mathbf{R}_l^{\mathrm{op}}\right)^{\mathrm{H}}\mathbf{H}\boldsymbol{\mu}. \tag{14b}$$

For massive MIMO systems, as the antenna number of the BS and the UT number rise, the computational complexity of (14) will become prohibitively high due to the large-dimensional matrix inversion and multiplication. To this end, we turn to exploring intrinsic structures of the optimal solution of (13).

Define the beam set and the beam selection matrix for group l as $\mathcal{B}_l \triangleq \cup_{u\in\mathcal{N}_l}\mathcal{A}_u$ and $\mathbf{B}_l \triangleq \mathbf{S}(A,\mathcal{B}_l) \in \mathbb{R}^{A\times B_l}$, respectively, where $B_l = |\mathcal{B}_l|$, and assume that the UT number is limited and the directional cosines of the paths from all UTs are distinct [4], then we have the following theorem. The proof is omitted due to the space limitation.

Theorem 1 *When $\forall l \neq l',\ \mathcal{B}_l \cap \mathcal{B}_{l'} = \varnothing$, as $M \to \infty$, the optimal solution of* (13) *can be obtained as*

$$\mathbf{R}_l^{\mathrm{op}} = \mathbf{V}\mathbf{B}_l\mathbf{W}_l^{\mathrm{op}}, \tag{15a}$$

$$\mathbf{d}_l^{\mathrm{op}} = \boldsymbol{\mu}_l - \left(\mathbf{W}_l^{\mathrm{op}}\right)^{\mathrm{H}}\mathbf{B}_l^{\mathrm{T}}\mathbf{G}\boldsymbol{\mu}, \tag{15b}$$

where $\boldsymbol{\mu}_l \triangleq \mathbf{N}_l^{\mathrm{T}}\boldsymbol{\mu} \in \mathbb{C}^{B_l}$, $\mathbf{\Sigma}_l \triangleq \mathbf{N}_l^{\mathrm{T}}\mathbf{\Sigma}\mathbf{N}_l \in \mathbb{R}^{B_l\times B_l}$, and

$$\mathbf{W}_l^{\mathrm{op}} = \arg\min_{\mathbf{W}_l\in\mathbb{C}^{B_l\times N_l}} \mathrm{tr}\Big\{\mathbf{W}_l^{\mathrm{H}}\left(\mathbf{B}_l^{\mathrm{T}}\mathbf{G}\mathbf{\Sigma}\mathbf{G}^{\mathrm{H}}\mathbf{B}_l + \sigma_{\mathrm{z}}\mathbf{I}\right)\mathbf{W}_l - \mathbf{\Sigma}_l\mathbf{N}_l^{\mathrm{T}}\mathbf{G}^{\mathrm{H}}\mathbf{B}_l\mathbf{W}_l - \mathbf{W}_l^{\mathrm{H}}\mathbf{B}_l^{\mathrm{T}}\mathbf{G}\mathbf{N}_l\mathbf{\Sigma}_l + \mathbf{\Sigma}_l\Big\}. \tag{16}$$

Theorem 1 reveals that under the criterion of minimizing MSE, the asymptotically optimal space domain detector is beam structured. Although the number of BS antennas is finite in practical systems, when M is sufficiently large, Theorem 1 can still guide the detector design. Specifically, for group l, we restrict the space domain detector $\mathbf{R}_l$ as beam structure

$$\mathbf{R}_l = \mathbf{V}\mathbf{B}_l\mathbf{W}_l, \tag{17}$$

where $\mathbf{W}_l \in \mathbb{C}^{B_l\times N_l}$ is the beam domain detector for group l. Combining (12) and (17), the MSE for the detection of $\mathbf{x}_l, l \in \mathcal{Z}_L^+$ is given by

$$\bar{\epsilon}_l(\mathbf{W}_l, \mathbf{d}_l) = \mathbb{E}\left\{\left\|\mathbf{x}_l - \left(\mathbf{W}_l^{\mathrm{H}}\mathbf{y}_l + \mathbf{d}_l\right)\right\|_2^2\right\}, \tag{18}$$

where $\mathbf{y}_l \triangleq \mathbf{B}_l^{\mathrm{T}}\mathbf{V}^{\mathrm{H}}\mathbf{y} \in \mathbb{C}^{B_l}$. By minimizing $\bar{\epsilon}_l(\mathbf{W}_l, \mathbf{d}_l)$, the detector for group l can be obtained as [21]

$$\begin{aligned}\mathbf{W}_l &= \mathrm{Cov}\{\mathbf{y}_l,\mathbf{y}_l\}^{-1}\mathrm{Cov}\{\mathbf{y}_l,\mathbf{x}_l\}\\ &= \left(\mathbf{D}_l\mathbf{\Sigma}\mathbf{D}_l^{\mathrm{H}} + \sigma_{\mathrm{z}}\mathbf{B}_l^{\mathrm{T}}\mathbf{Q}\mathbf{B}_l\right)^{-1}\widetilde{\mathbf{D}}_l\mathbf{\Sigma}_l, \qquad (19a)\\ \mathbf{d}_l &= \mathbb{E}\{\mathbf{x}_l\} - \mathrm{Cov}\{\mathbf{x}_l,\mathbf{y}_l\}\mathrm{Cov}\{\mathbf{y}_l,\mathbf{y}_l\}^{-1}\mathbb{E}\{\mathbf{y}_l\}\\ &= \boldsymbol{\mu}_l - \mathbf{W}_l^{\mathrm{H}}\mathbf{D}_l\boldsymbol{\mu}, \qquad (19b)\end{aligned}$$

where $\mathbf{D}_l \triangleq \mathbf{B}_l^{\mathrm{T}}\mathbf{D} \in \mathbb{C}^{B_l\times U}$, $\widetilde{\mathbf{D}}_l \triangleq \mathbf{D}_l\mathbf{N}_l \in \mathbb{C}^{B_l\times N_l}$, and

$$\mathbf{D} \triangleq \mathbf{Q}\mathbf{G} \in \mathbb{C}^{A\times U}, \tag{20a}$$

$$\mathbf{Q} \triangleq \mathbf{V}^{\mathrm{H}}\mathbf{V} \in \mathbb{C}^{A\times A}. \tag{20b}$$

Consequently, the detection for group l is given as

$$\widehat{\mathbf{x}}_l = \mathbf{W}_l^{\mathrm{H}}\mathbf{y}_l + \mathbf{d}_l = \mathbf{W}_l^{\mathrm{H}}\widetilde{\mathbf{y}}_l + \boldsymbol{\mu}_l, \tag{21}$$

where $\widetilde{\mathbf{y}}_l \triangleq \mathbf{B}_l^{\mathrm{T}}\widetilde{\mathbf{y}} \in \mathbb{C}^{B_l}$ is a subvector of $\widetilde{\mathbf{y}} = \mathbf{V}^{\mathrm{H}}\mathbf{y} - \mathbf{D}\boldsymbol{\mu}$.

In practice, to achieve satisfactory performance with relatively low complexity, appropriate UT scheduling and grouping strategies can be employed to minimize the cardinality of the beam set of each UT group. Given the pronounced sparsity of the beam domain channel and the grouping scheme, the computation of (19a) and (21), the so-called beam domain detector design and beam structured signal detection, only involves low-dimensional operations, resulting in relatively low complexity. Moreover, the beam structured signal detection in [16] can be viewed as a special case of the proposed grouping-based iterative approach, namely, when each group contains only one UT and there is no iterative update of the a priori information.

After the beam structured signal detection using the a priori information, the extrinsic information calculation can be effectively implemented. Assuming $\mathbf{x} \sim \mathcal{CN}(\boldsymbol{\mu}, \mathbf{\Sigma})$, for $u \in \mathcal{N}_l,\ l \in \mathcal{Z}_L^+$, we have [22]

$$p(\mathbf{y}_l|x_u) = \frac{p(x_u|\mathbf{y}_l)p(\mathbf{y}_l)}{p(x_u)} \propto \exp\left\{-\frac{|x_u - \mu_u^{\mathrm{e}}|^2}{\sigma_u^{\mathrm{e}}}\right\}, \tag{22}$$

where the extrinsic mean and variance of x_u are expressed as

$$\mu_u^{\mathrm{e}} = (\mu_u^{\mathrm{p}}/\sigma_u^{\mathrm{p}} - \mu_u/\sigma_u)\sigma_u^{\mathrm{e}}, \tag{23a}$$

$$\sigma_u^{\mathrm{e}} = (1/\sigma_u^{\mathrm{p}} - 1/\sigma_u)^{-1}, \tag{23b}$$

respectively, and

$$\mu_u^{\mathrm{p}} = \mathbb{E}\{x_u|\mathbf{y}_l\} = \mathbf{e}_u^{\mathrm{T}}\mathbf{N}_l\widehat{\mathbf{x}}_l, \tag{24a}$$

$$\sigma_u^{\mathrm{p}} = \mathrm{Cov}\{x_u, x_u|\mathbf{y}_l\} = \left(1 - \mathbf{e}_u^{\mathrm{T}}\mathbf{N}_l\mathbf{W}_l^{\mathrm{H}}\mathbf{D}_l\mathbf{e}_u\right)\sigma_u, \tag{24b}$$

are denoted as the a posteriori mean and the a posteriori variance of x_u, respectively. By approximating $p(\mathbf{y}|x_u)$ as $p(\mathbf{y}_l|x_u)$, the extrinsic LLR $L_{\mathrm{e}}(q_{u,i})$ can be computed by

$$L_{\mathrm{e}}(q_{u,i}) = \ln\frac{\sum_{s_k\in\mathbb{S}_i^1}\exp\left\{-|\mu_u^{\mathrm{e}} - s_k|^2/\sigma_u^{\mathrm{e}} + \sum_{j\neq i}^{N}(b_{k,j} - 1/2)L_{\mathrm{a}}'(q_{u,j})\right\}}{\sum_{s_k\in\mathbb{S}_i^0}\exp\left\{-|\mu_u^{\mathrm{e}} - s_k|^2/\sigma_u^{\mathrm{e}} + \sum_{j\neq i}^{N}(b_{k,j} - 1/2)L_{\mathrm{a}}'(q_{u,j})\right\}}.$$

In addition to the beam structured signal detection and the extrinsic information calculation, the a priori information update stays consistent with that described in III-A, thereby forming the beam structured turbo receiver (BSTR).

C. Efficient Implementation of BSTR

In order to efficiently implement the BSTR, we discuss the low-complexity computations of $\widetilde{\mathbf{y}}$ and $\mathbf{D}$ for $\widetilde{\mathbf{y}}_l$ and $\mathbf{D}_l$ in (21) in conjunction with their structural properties.

The computation of $\widetilde{\mathbf{y}}$ is required to be implemented at each turbo iteration since the a priori mean $\boldsymbol{\mu}$ is updated accordingly. Exploiting the structure of the beam matrix (7) and combining $\left(\overline{\mathbf{\Pi}}_S^N\right)^{\mathrm{T}} = -\overline{\mathbf{\Pi}}_S^{S-N}$, we have

$$\begin{aligned}\widetilde{\mathbf{y}} =& \mathbf{V}^{\mathrm{H}}\left(\mathbf{y} - \mathbf{V}\mathbf{G}\boldsymbol{\mu}\right)\\ =& -\frac{1}{M}\mathbf{I}_{A,S}\overline{\mathbf{\Pi}}_S^{(A-1)/2}\mathbf{\Omega}^*\mathbf{F}_S^{\mathrm{H}}\mathbf{I}_{S,M}\\ &\times\left(\sqrt{M}\mathbf{y} - \mathbf{I}_{M,S}\mathbf{F}_S\mathbf{\Omega}\overline{\mathbf{\Pi}}_S^{S-(A-1)/2}\mathbf{I}_{S,A}\mathbf{G}\boldsymbol{\mu}\right). \qquad (25)\end{aligned}$$

Then the computation of $\widetilde{\mathbf{y}}$ can be efficiently implemented since $\mathbf{G}$ is sparse and fast Fourier transform (FFT) can be adopted.

The computation of $\mathbf{D}$ is required to be implemented at each channel realization since it only varies with $\mathbf{G}$. By substituting (7), $\mathbf{D}$ can be rewritten as $\mathbf{D} = \mathbf{Q}\mathbf{G} = \mathbf{I}_{A,S}\widetilde{\mathbf{Q}}\mathbf{I}_{S,A}\mathbf{G}$, where

$$\begin{aligned}\widetilde{\mathbf{Q}} &= -\frac{1}{M}\overline{\mathbf{\Pi}}_S^{(A-1)/2}\mathbf{\Omega}^*\mathbf{F}_S^{\mathrm{H}}\mathbf{I}_{S,M}\mathbf{I}_{M,S}\mathbf{F}_S\mathbf{\Omega}\overline{\mathbf{\Pi}}_S^{S-(A-1)/2} \\ &= -\frac{1}{M}\overline{\mathbf{\Pi}}_S^{(A-1)/2}\left(M\sum_{k=0}^{S-1}\gamma_k\overline{\mathbf{\Pi}}_S^k\right)\overline{\mathbf{\Pi}}_S^{S-(A-1)/2} \\ &= -\sum_{k=0}^{S-1}\gamma_k\overline{\mathbf{\Pi}}_S^{(A-1)/2}\overline{\mathbf{\Pi}}_S^k\overline{\mathbf{\Pi}}_S^{S-(A-1)/2} = \sum_{k=0}^{S-1}\gamma_k\overline{\mathbf{\Pi}}_S^k, \quad (26)\end{aligned}$$

in which $\gamma_k = \sin(\pi k/F)/(M\sin(\pi k/S)) \in \mathbb{R},\ k \in \mathbb{Z}$. It can be checked that $\gamma_k = (-1)^{M+1}\gamma_{S-k},\ k \in \mathbb{Z}$, thus $\widetilde{\mathbf{Q}}$ can be rewritten as $\widetilde{\mathbf{Q}} = \mathbf{I}_S + \sum_{k=1}^{\lceil S/2\rceil-1}\gamma_k\left(\overline{\mathbf{\Pi}}_S^k + (-1)^{M+1}\overline{\mathbf{\Pi}}_S^{S-k}\right)$. Consequently, $\mathbf{D}$ can be expressed as

$$\begin{aligned}\mathbf{D} &= \mathbf{I}_{A,S}\widetilde{\mathbf{Q}}\mathbf{I}_{S,A}\mathbf{G} \\ &= \mathbf{G} + \underbrace{\mathbf{I}_{A,S}\sum_{k=1}^{\lceil S/2\rceil-1}\left(\overline{\mathbf{\Pi}}_S^k + (-1)^{M+1}\overline{\mathbf{\Pi}}_S^{S-k}\right)\mathbf{I}_{S,A}\mathbf{G}\gamma_k}_{\triangleq\mathbf{G}_{\mathrm{d}}},\end{aligned} \quad (27)$$

where $\mathbf{G}_{\mathrm{d}}$ can be viewed as diffusion of the beam domain channel $\mathbf{G}$. From (27), the computation of $\mathbf{D}$ arises from the computation of $\mathbf{G}_{\mathrm{d}}$, involving the multiplications of a sparse complex-valued matrix and real-valued scalars (i.e., $\mathbf{G}\gamma_k,\ k \in \mathcal{Z}^+_{\lceil S/2\rceil-1}$), along with several permutations, sign inversions, extractions, and summation operations.

D. Complexity Analysis

We evaluate the computational complexity of the proposed BSTR by counting the number of complex multiplications (CMs), where the multiplication between a real scalar and a complex scalar is viewed as half CM. To better examine the computational complexity of turbo receivers, only the complexity of signal detection is considered since it is dominant, while the complexities of other modulos (e.g., the extrinsic information calculation, the a priori information update and decoding) are not taken into consideration. The number of iterations is denoted as T. Average values $\widetilde{A} \triangleq \frac{1}{U}\sum_{u=1}^{U}A_u$ and $B \triangleq \frac{1}{L}\sum_{l=1}^{L}B_l$ are defined for the convenience of complexity analysis.

To begin with, we analyze the computational complexity of the BSTR. Before the iteration, $\mathbf{D}$ needs to be computed, which requires $\widetilde{A}U(\lceil S/2\rceil - 1)/2$ CMs by (27). The complexity for each iteration comes from the computation of $\widehat{\mathbf{x}}_l,\ l \in \mathcal{Z}_L^+$, where the computation of $\widetilde{\mathbf{y}}$ can be efficiently implemented by (25) with $\widetilde{A}U + A + S(1+\log S)$ CMs, and the computation of (19a) and (21) consumes $(B(B+2)U + B^2(B+3))L/2 + B(B+1)U$ CMs. Therefore, the total complexity of the BSTR is $\widetilde{A}U(\lceil S/2\rceil - 1)/2 + (\widetilde{A}U + A + S(1+\log S) + (B(B+2)U + B^2(B+3))L/2 + B(B+1)U)T$.

The computational complexity of the MMSE turbo receiver is also included as a benchmark. Before the iteration, $\mathbf{H}^{\mathrm{H}}\mathbf{H}$ is computed with $MU(U+1)/2$ CMs. At each iteration, the complexity of signal detection is given by $(U+5)U^2/2 + MU(U+2)$. Then the complexity of the MMSE turbo receiver is $MU(U+1)/2 + ((U+5)U^2/2 + MU(U+2))T$ in total. In massive MIMO systems, when M and U are both large, the proposed BSTR has a markedly lower complexity in comparison to the MMSE turbo receiver.

IV. Simulation Results

In this section, simulation results of the proposed receiver for HF skywave massive MIMO systems are provided.

The simulation parameters are presented in Table I. UTs are distributed in a 140° sector around 2000 km from the BS with $L = 18$ groups, and UT scheduling is performed to ensure that UTs from different groups exhibit relatively low channel correlation. Channel parameters are generated by the ray-tracing software Proplab-Pro 3.1 [15]. The modulation scheme is 64 quadrature amplitude modulation (QAM), and the channel code is selected as low density parity check (LDPC) code of length 2112 with code rate $3/4$.

TABLE I
Simulation Parameters

Parameter	Value
Carrier frequency f_{c}	16 MHz
Number of antennas at the BS M	256
Antenna spacing at the BS d	9 m
Fine factor F	2
Number of UTs U	72

We adopt the bit error rate (BER) as the performance metric, and the signal-to-noise ratio (SNR) refers to the received SNR. Specifically, the following receivers are compared:

- **BSTR:** Our proposed beam structured turbo receiver.
- **MMSE TR:** MMSE turbo receiver [21].
- **BSD:** Beam structured detector [16].
- **JSTD:** Joint Slepian transform based detector [15].

Fig. 1 provides the BER performance of the BSTR, compared with the MMSE TR, BSD and JSTD. The number of turbo iterations is set as 3. The performance of the BSTR is approaching that of the MMSE TR at each iteration, especially for the final iteration. Moreover, the BSTR outperforms the BSD and JSTD at the first iteration, and the performance gap is enlarged during the subsequent iterations. Meanwhile, after the first iteration, the performance of the BSTR is remarkably superior to the first iteration of the MMSE TR (i.e., the MMSE detection without the a priori information update).

In Fig. 2, the computational complexity of the BSTR is depicted versus the number of UTs, together with the MMSE TR, BSD and JSTD. The complexity of the BSTR with 3 iterations is close to that of the BSD and lower than that of the MMSE TR with even 1 iteration, while it is higher than that of the JSTD when the number of UT is large. Nevertheless, compared to the other receivers, the BSTR achieves a superior performance with a relatively low complexity.

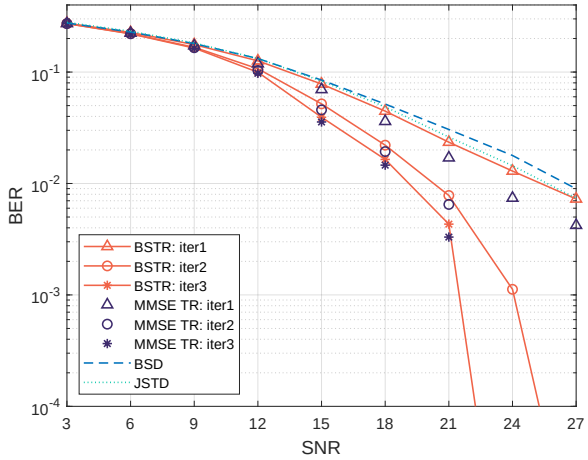


Fig. 1. BER performance of the BSTR, compared with the MMSE TR, BSD and JSTD.

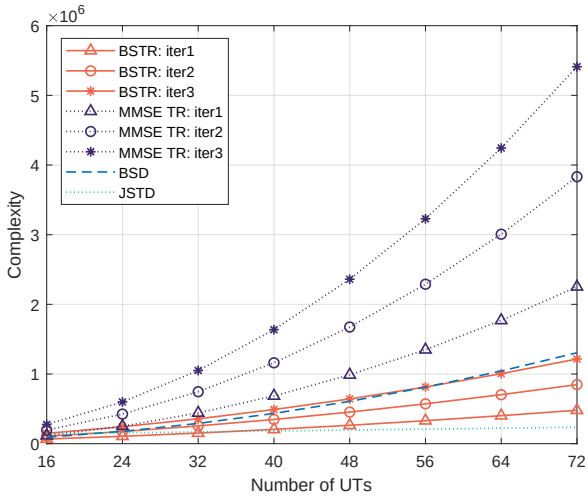


Fig. 2. Computational complexity of the BSTR, compared with the MMSE TR, BSD and JSTD.

V. Conclusions

In this paper, we investigated receiver design for HF skywave massive MIMO systems. We first proposed a modified beam based channel model and constructed the beam matrix as DFT-based structure for low-complexity receiver design. Then, we proposed the BSTR, involving low-dimensional beam structured signal detection for each UT group. Additionally, efficient Implementation of the BSTR was discussed by using the characteristics of the beam matrix and the beam domain channel sparsity. Finally, the effectiveness of the proposed BSTR was demonstrated by simulation results.

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